SentinelQ

An automated signal engine for portfolio monitoring — executive brief on methodology, reliability and production readiness

SentinelQ reads the public news, disclosures and corporate-action history of every stock in a portfolio and turns them into three independent, dated, source-linked scores: how the market narrative is trending, whether anything adverse has happened in governance, and how much risk is sitting in recent corporate actions. It is built to be checked, not trusted blindly — every number traces back to a specific article and a fixed, published rule, and the same inputs always produce the same output. This brief summarises what it does, why its output can be relied on, what has been built and tested so far, and what remains before it is a production system of record rather than a prototype.

What it produces

For every stock in a portfolio, SentinelQ scores three separate dimensions, each on its own fixed scale:

Dimension	Scale	What it captures
Sentiment	-2 to +2	How the past year of reported news reads for the company — results, guidance, orders, analyst views — weighted so recent news counts more than old news. Price moves never feed this score; only reported, dated events do.
Governance	0 to 100	Starts clean and loses fixed points only for specific, verified adverse events (a regulator's order, an auditor's exit, an unplanned senior departure, a promoter pledge being called). Routine events — a scheduled retirement, a resolved inquiry — are reviewed and explicitly not penalised.
Corporate Actions	0 to 10	How much risk is concentrated in a stock's recent capital, ownership or structural activity — a dilutive raise, a change of control, a promoter pledge — combined so one serious event is never diluted by several minor ones.

A fourth, read-only layer — dividends, buybacks, splits and similar mechanical actions — is listed for context but deliberately kept out of all three scores: these do not change what a company is worth, and including them would distort the signal rather than inform it.

What specifically gets marked

The three dimensions are not a vague sentiment reading — each is built from a fixed, named list of things the system watches for. What follows is that list in plain terms, so it is clear exactly what drives a score up or down rather than treating it as a black box.

Sentiment — what counts as strongly positive through strongly negative

Reading	What produces it
Strongly positive	A sustained run of results beating expectations, raised guidance, or a structural tailwind, with no material offsetting bad news in the period.
Positive on balance	A good year with one specific qualifier — a single weak quarter, an execution issue, or a valuation caution — that keeps it short of the strongest reading.
Neutral / mixed	Genuinely two-sided — real good news matched by real bad news — or coverage too thin to call a direction with confidence. A thin-coverage neutral is flagged as such, not presented as a confident zero.

Reading	What produces it
Negative on balance	A material earnings deterioration, a guidance cut, or a defined adverse event whose impact dominates the period, even where some offsetting positives exist.
Strongly negative	Severe or compounding negatives — a large miss stacked with further bad news, not one isolated setback.

Governance — the specific events that can cost points, and what does not

Severity	Examples
Most severe	A financial restatement; an auditor resigning; the company entering formal insolvency proceedings or facing a long trading suspension; a promoter's pledged shares being forcibly sold into the market.
Serious	A regulator's final order or settlement (not a mere allegation); a live, unresolved investigation; a material, investor-contested related-party transaction.
Moderate	An unplanned departure of a CEO, CFO, company secretary or other senior compliance role; a gap in board independence (combined chair/MD roles, too few independent directors); a promoter pledge being created at a size judged material.
Minor / procedural	A disclosure lapse or an exchange fine for a procedural breach, scaled by its size — never treated as an integrity matter.

Reviewed and explicitly NOT marked down: a scheduled leadership retirement or succession, an ordinary board reconstitution, a routine or arm's-length related-party transaction, an investigation that was closed or cleared, a scheduled auditor rotation. Each of these is checked against the same list and recorded as “reviewed, not penalised” rather than silently ignored — the distinction between a real governance concern and routine corporate housekeeping is deliberate, not an oversight.

Corporate Actions — the full range tracked, by severity

Severity	Examples
Most severe	Insolvency proceedings or compulsory delisting; a demerger or a merger that changes what the business is.
High	A takeover or open offer that changes who controls the company; a stock split or bonus issue (tracked for its effect on reported data, not treated as a risk event in itself).
Material	A rights issue; a promoter pledge being created or called; a dilutive capital raise (a preferential issue, a institutional placement, or similar); an auditor or senior-management exit (counted here as well as under Governance, since the same event carries both a governance and a capital-markets dimension).
Moderate	A buyback; a lock-in expiry that will release a block of shares for sale.
Routine	A regular dividend; a name, symbol or ISIN change; a change of financial year-end; a routine AGM/EGM resolution; administrative notices (registrar changes, trading-window closures) that carry essentially no signal on their own.

A small number of event types are tracked on the calendar for completeness (for example an index addition or removal, or an exchange surveillance action) but deliberately excluded from the score itself, since the rules distinguish a formal corporate action from a market-structure disclosure.

How it works

The system follows a fixed, six-stage pipeline. Only one stage uses AI, and its role is deliberately narrow: it reads one news article at a time and assigns it a label from a closed, pre-defined list — an event type, a sentiment value on a fixed scale, and whether it concerns governance. It never computes a score, never sees

the scoring rules, and never aggregates anything across articles. Every other stage — retrieving the news, filtering out noise, turning labels into scores, writing the report — is ordinary, deterministic software with no AI involved and no randomness.

The labelling step recognises 45 distinct event types spanning results, governance, and the full range of formal corporate actions (capital raises, mergers, buybacks, pledges, insolvency and more), each mapped to a documented severity and treatment.

Stage	What happens
1. Input	Resolve the exact company identity (name, ticker, sector) so every later search is unambiguous.
2. Collect	Retrieve news, results coverage and corporate-action history for the lookback period; discard templated and off-topic items before anything is read in depth; merge duplicate coverage of the same story into one event so repetition is never counted as repeated evidence.
3. Label	The one AI step: read each distinct event and assign it a fixed-vocabulary label — nothing else.
4. Validate	Check every label against the expected structure; anything malformed is retried once, then dropped with a recorded reason — never silently kept or silently discarded.
5. Score	Deterministic arithmetic only. Turn labels into the three scores using fixed, published rules — no AI, no judgement calls, fully reproducible by hand from the same inputs.
6. Report	Produce the scorecard and the full supporting evidence — every score traceable to the specific, dated, linked articles behind it.

How a score is actually built

The arithmetic is simple on purpose, and it is worth showing directly rather than asserting it is sound. Two illustrations, using round hypothetical numbers — not a real company:

Governance: a running total, not a rolling average

Every company starts at 100. A verified adverse event subtracts a fixed amount set in advance by its type — an auditor's resignation costs 20 points, a regulator's settled order costs 20, a procedural exchange fine costs 5 to 10 depending on its size. Say a company has a regulator's order (-20) and, eight months later, an unplanned CFO exit (-8): the running total is 100 - 20 - 8 = 72. Old penalties do not vanish, but they do fade — a penalty holds at full strength for six months, then tapers in a straight line to half its value by twelve months, so an integrity event from last year still shows, just more lightly than one from last month. Nothing is estimated or smoothed; the number is always exactly '100, minus this list of dated events, each worth a fixed, published amount.'

Corporate actions: one serious event is never diluted by several small ones

Each event gets a 0-10 score built from four factors multiplied together: how dangerous that type of event is in the worst case, how far along it is (a board merely discussing a capital raise counts for far less than one that has formally approved it), how recent it is, and how large this instance is. Say a company in a given year has a capital raise scored 6.5 and a routine administrative notice scored 0.3. These are not averaged — averaging would let the 0.3 water down the 6.5 — and they are not added — summing lets unrelated minor items stack past any sensible ceiling. Instead each is treated as a chance out of ten that it causes real harm, and the two are combined as the chance that at least one of them does: here, that works out to a combined score of 6.6 out of 10 — barely above the single largest event, exactly as it should be, because one real risk dominates and a minor administrative item adds almost nothing.

Both calculations are reproducible by hand from the published rules and the dated evidence list behind them — there is no step where a model estimates, smooths, or rounds a result.

Why the output can be trusted

Full traceability. Every score is backed by the specific, dated articles that drove it, each with its own individual assessment and a working link. Any score can be opened down to the evidence behind it.

Source-strength is shown, not assumed. Evidence is weighted by how many outlets reported it and how reliable those outlets are. A claim resting on one low-reliability source is flagged as such in the output, not presented with the same confidence as a well-corroborated one.

Verification before penalty. An adverse governance event is checked to confirm it concerns the company itself — not a peer, a counterparty, or an industry-wide rule sharing a keyword — before it can cost a point. Items that fail this check are logged as reviewed and excluded, not dropped silently.

Failures are recorded, never hidden. An unreadable article, a malformed label, an unresolved stock name — each produces a recorded reason. A run with too many failures stops rather than publishing on shaky evidence.

What has been built and tested

The current system is a working prototype, not a paper design: it runs end to end against live, public news sources and produces the full scorecard and supporting workbook for any named list of stocks today. The deterministic scoring logic — the part of the system that turns evidence into a number, and the part that matters most for consistency and auditability — is covered by an automated test suite of 166 checks that run before any change is accepted, each one pinned to a specific, hand-verified expected result. The system has been run and reviewed against real portfolios of varying size, with its output checked line by line against the underlying articles.

What production readiness would add

The prototype demonstrates the methodology works. Moving it to a system the firm relies on day to day would mean addressing the following, in roughly the order they matter:

Area	What it would mean
Primary-source data	Today's evidence is drawn from public news coverage. Direct integration with exchange filings and regulatory disclosures would reduce dependence on how well a given event happened to be reported, and is the single highest-value addition for governance and corporate-action coverage specifically.
Human sign-off workflow	A defined point at which a person reviews and approves scores before they reach decision-makers — particularly for the lowest-scoring names and anything a verification check could not fully resolve on its own.
Scheduled, monitored operation	Running on a fixed schedule with alerting if a run fails or produces an unexpected shift, rather than being run on request.
Coverage breadth and calibration	Testing across a wider and more varied set of names than has been exercised so far, to confirm the scoring thresholds hold up across sectors and company sizes, with a feedback loop to tune them as real outcomes come in.
Access and distribution	A defined way for the people who need these scores to receive them — a shared location, a scheduled distribution, or integration into an existing review process.

Recommendation

The methodology is sound, deterministic, and fully auditable, and the prototype already produces usable output against live data. The most efficient next step is a short pilot against a defined, representative portfolio, with a named reviewer validating a sample of scores against source filings before anything from the system reaches a wider audience. That pilot would surface any remaining gaps directly, rather than theorising about them, and gives a concrete basis for deciding the investment in the production items above.

This is a summary for decision-making purposes. The full technical reference — every scoring rule, formula and worked example — is available on request.